

CA2 REPORT: ENVIRONMENTAL SCIENCE

Student Name: [Your Name]

Subject: Environmental Science / Engineering

Assignment: Continuous Assessment 2 (CA2)

QUESTION 1

(a) Explain the term NTLV. State its significance. State the condition where the term NTLV is used to define noise pollution level in a given area.

Definition: NTLV stands for **Noise Threshold Limit Value**. It represents the maximum permissible sound pressure level (noise level) to which a person can be exposed for a specific duration (usually a standard 8-hour workday) without risking permanent hearing damage or adverse health effects.

Significance:

1. **Occupational Safety:** It serves as a critical benchmark for industrial safety, helping to define "safe" working environments.
2. **Regulatory Compliance:** It provides a legal standard for environmental protection agencies to monitor and penalize noise violations.
3. **Health Protection:** It prevents Noise-Induced Hearing Loss (NIHL) and non-auditory effects like stress and hypertension.

Condition for Defining Noise Pollution: The term NTLV is used to define a state of **noise pollution** when the measured **Equivalent Continuous Noise Level (L_{eq})** in a given area exceeds the prescribed Threshold Limit Value for that specific zone (e.g., Residential, Commercial, or Industrial) over a specified time period.

- *Example:* If the NTLV for a residential zone is 55 dB (Daytime) and the measured level is 65 dB, the condition is defined as Noise Pollution.

(b) In a given area two noise levels, 90 dB and 120 dB are active. Calculate the intensity ratio of the two noise levels.

Formula: Difference in Sound Level (ΔL):

$$\Delta L = L_2 - L_1$$

Intensity Ratio Formula:

$$\frac{I_2}{I_1} = 10^{\frac{\Delta L}{10}}$$

Given Values & Constants:

- $L_1 = 90 \text{ dB}$
- $L_2 = 120 \text{ dB}$

Calculation:**1. Calculate Difference:**

$$\Delta L = 120 - 90 = 30 \text{ dB}$$

2. Calculate Ratio:

$$\frac{I_2}{I_1} = 10^{\frac{30}{10}}$$

$$\frac{I_2}{I_1} = 10^3$$

$$\frac{I_2}{I_1} = 1000$$

Result: The intensity of the 120 dB sound is **1000 times** greater than the 90 dB sound.

(c) Calculate the average of the following noise levels: 70 dB, 58 dB, 62 dB, 80 dB, 49 dB, 76 dB.

Formula: Logarithmic Average (L_{avg}):

$$L_{avg} = 10 \log_{10} \left[\frac{1}{N} \sum_{i=1}^N 10^{\frac{L_i}{10}} \right]$$

Where:

- N = Total number of readings
- L_i = Individual noise levels

Given Values:

- $L_1 = 70, L_2 = 58, L_3 = 62, L_4 = 80, L_5 = 49, L_6 = 76$
- $N = 6$

Calculation:**1. Convert dB to Intensity Factors ($10^{L/10}$):**

- $10^{7.0} = 10,000,000$
- $10^{5.8} \approx 630,957$
- $10^{6.2} \approx 1,584,893$
- $10^{8.0} = 100,000,000$

- $10^{4.9} \approx 79,433$
 - $10^{7.6} \approx 39,810,717$
2. **Summation (Σ):** Sum $\approx 152,106,000$
 3. **Average Intensity:**

$$\text{Avg} = \frac{152,106,000}{6} \approx 25,351,000$$

4. **Convert back to dB:**

$$L_{avg} = 10 \log_{10}(25,351,000)$$

$$L_{avg} \approx 74.03 \text{ dB}$$

Result: 74.03 dB

QUESTION 2

(a) Discuss in brief adverse effects of noise pollution.

The adverse effects of noise pollution are generally classified into two categories:

1. **Auditory Effects (Direct impact on hearing):**

- **Auditory Fatigue:** Temporary shifting of the hearing threshold after exposure to loud noise (e.g., ringing ears after a concert).
- **Deafness/Hearing Loss:** Can be temporary or permanent. Chronic exposure to noise >90 dB destroys the hair cells in the cochlea (Organ of Corti), leading to Noise-Induced Hearing Loss (NIHL).

2. **Non-Auditory Effects (Impact on general health):**

- **Physiological:** Increases heart rate, blood pressure (hypertension), and risk of cardiovascular disease. It also causes pupil dilation and digestive spasms.
- **Psychological:** Causes annoyance, stress, anxiety, irritability, and sleep disturbances (insomnia).
- **Performance:** Reduces concentration and efficiency in tasks requiring focus.
- **Communication:** Masks speech, causing misunderstandings and social isolation.

(b) What is called Lepn noise index? Where is it used?

Definition: L_{epn} (or **EPNL**) stands for **Effective Perceived Noise Level**. It is a complex noise rating unit that accounts for:

1. The maximum noise level during an event.
2. The duration of the noise event.

3. The presence of discrete frequencies (pure tones), which are judged to be more annoying than broadband noise.

Usage:

- **Aircraft Noise Certification:** It is primarily used by the International Civil Aviation Organization (ICAO) and Federal Aviation Administration (FAA) for certifying aircraft noise levels.
- **Airport Zoning:** It is used to monitor noise around airports to assess the impact of single-event aircraft flyovers on the surrounding community.

(c) Calculate Leq of that area.

Formula: Equivalent Continuous Level (L_{eq}):

$$L_{eq} = 10 \log_{10} \left[\frac{1}{T} \sum_{i=1}^n (t_i \times 10^{\frac{L_i}{10}}) \right]$$

Where:

- T = Total time duration ($\sum t_i$)
- t_i = Duration of specific noise
- L_i = Specific noise level

Given Data:

- (a) 87 dB for 67 min ($t_1 = 67, L_1 = 87$)
- (b) 90 dB for 12 min ($t_2 = 12, L_2 = 90$)
- (c) 80 dB for 92 min ($t_3 = 92, L_3 = 80$)
- (d) 76 dB for 124 min ($t_4 = 124, L_4 = 76$)

Calculation:

1. **Calculate Total Time (T):**

$$T = 67 + 12 + 92 + 124 = 295 \text{ min}$$

2. **Calculate Weighted Energies ($t_i \times 10^{L_i/10}$):**

- $67 \times 10^{8.7} = 67 \times 5.01 \times 10^8 \approx 335.79 \times 10^8$
- $12 \times 10^{9.0} = 12 \times 10 \times 10^8 = 120.00 \times 10^8$
- $92 \times 10^{8.0} = 92 \times 1.0 \times 10^8 = 92.00 \times 10^8$
- $124 \times 10^{7.6} = 124 \times 0.398 \times 10^8 \approx 49.36 \times 10^8$

3. **Summation:**

$$\sum \approx (335.79 + 120.00 + 92.00 + 49.36) \times 10^8$$

$$\sum \approx 597.15 \times 10^8$$

4. Average and Log:

$$\text{Average Energy} = \frac{597.15 \times 10^8}{295} \approx 2.024 \times 10^8$$

$$L_{eq} = 10 \log_{10}(2.024 \times 10^8)$$

$$L_{eq} = 10 \times (0.306 + 8)$$

$$L_{eq} = 83.06 \text{ dB}$$

Result: 83.06 dB

QUESTION 3

Define ecological pyramids and describe with examples pyramid of number and pyramid of biomass.

Definition: An **Ecological Pyramid** is a graphical representation designed to show the biomass, bio-productivity, or number of organisms at each trophic level in a given ecosystem. The base of the pyramid always represents the producers (autotrophs).

1. Pyramid of Number:

- **Description:** It depicts the number of individual organisms at different trophic levels of the food chain.
- **Upright Example (Grassland Ecosystem):**
 - *Base:* Millions of grasses (Producers).
 - *Middle:* Thousands of insects/herbivores.
 - *Top:* Few carnivores (e.g., Hawks/Lions).
 - *Shape:* Upright.
- **Inverted Example (Tree Ecosystem):**
 - *Base:* One single large tree.
 - *Middle:* Many birds feeding on the tree.
 - *Top:* Numerous parasites on the birds.
 - *Shape:* Inverted (Spindle shaped).

2. Pyramid of Biomass:

- **Description:** It depicts the total dry weight (biomass) of organisms at each trophic level. It indicates the amount of living matter present.

- **Upright Example (Forest/Grassland):**
 - The total mass of trees/grass is huge.
 - The mass of herbivores is less.
 - The mass of top carnivores is the least.
- **Inverted Example (Aquatic/Pond Ecosystem):**
 - *Base:* Phytoplankton (tiny producers) have low biomass at any specific time but reproduce fast.
 - *Top:* Large fishes (consumers) have a much larger accumulated biomass.
 - *Shape:* Inverted.

QUESTION 4

Discuss biological magnification. Will the levels of this magnification be different at different levels of the ecosystem?

Biological Magnification (Biomagnification): Biological magnification refers to the phenomenon where the concentration of non-degradable pollutants (such as DDT, mercury, or heavy metals) increases progressively at successive trophic levels of a food chain.

Because these toxins cannot be metabolized or excreted by organisms, they accumulate in the fatty tissues. When a consumer eats many such organisms, they absorb the total toxin load of their prey.

Different Levels of Magnification: Yes, the levels of magnification will be different at different levels of the ecosystem.

- **Mechanism:** The concentration increases significantly as we move **up** the food chain.
- **Lower Trophic Levels (Producers):** Have the lowest concentration of the toxin (e.g., DDT in water = 0.003 ppb).
- **Middle Trophic Levels (Herbivores/Small Fish):** Have a moderate concentration (e.g., Zooplankton = 0.04 ppm).
- **Higher Trophic Levels (Top Carnivores):** Have the highest concentration. Since top carnivores consume large quantities of prey to sustain themselves, the toxins from all that